

DAX Functions

Function 1: SUM()

SUM() adds all the values in one numerical column.
It calculates the total of a column.

Syntax: SUM(TableName[ColumnName])

```
Total Sales:=SUM(Sales_Fact[Sales_Amount])
```

Use cases:

- Total Sales
- Total Profit
- Total Revenue
- Total Cost
- Total Quantity

Function 2: COUNTROWS()

COUNTROWS() counts how many rows exist in a table
It counts records

Syntax: COUNTROWS(TableName)

```
Total Orders:=COUNTROWS(Sales_Fact)
```

Use cases:

- How many orders?
- How many employees?
- How many customers?
- How many products?

Interview Question:

Difference between COUNT(), COUNTA() & COUNTROWS()?

Answer:

- COUNT() counts numeric values in a column.
- COUNTA() counts numeric, text, special chars (Non-Empty) in a column
- COUNTROWS() counts the total rows of a table.

Function 3: DISTINCTCOUNT()

DISTINCTCOUNT() counts only unique values and ignores duplicates
It counts the unique items.

Syntax: DISTINCTCOUNT(TableName[Column])

```
Unique Customers:=DISTINCTCOUNT(Sales_Fact[Customer_ID])
```

Use cases:

- Unique Customers
- Unique Products
- Unique Employees

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Unique Branches

Unique Cities

Interview Question:

Why not use COUNTROWS() here?

Answer:

Because one customer may appear many times/multiple orders.

DISTINCTCOUNT() remove duplicates and counts only unique values.

Function 4: DIVIDE()

DIVIDE() safely divides one number by another number

It performs division and prevents divide-by-zero errors.

Syntax: DIVIDE(Numerator,Denominator,[Alternate Result])

Profit Margin:=DIVIDE([Total Profit],[Total Sales])

Use cases:

Profit Margin

Average Price

Revenue Per Customer

Sales Per Employee

Cost Per Unit

Interview Question:

Why DIVIDE() preferred over "/" in DAX?

Because DIVIDE() safely handles divide by zero situations and returns a reliable result.

Function	Purpose	Example
SUM()	Adds values	Total Sales
COUNTROWS()	Counts records	Total Orders
DISTINCTCOUNT()	Counts unique values	Unique Customers
DIVIDE()	Safe division	Profit Margin

DAX Functions

ROW CONTEXT & FILTER CONTEXT

If you understand Context, 70% of DAX becomes easy

What is Context?

Context is the environment in which a DAX formula is evaluated.

Context tells DAX which data should be used while calculating.

There are two main contexts

1. Row Context
2. Filter Context

ROW Context

Row Context means DAX performs the calculation one row at a time

DAX looks at one record, calculates the result, then moves to the next record.

Order_ID	Sales	Profit	Profit %
1001	10000	2000	20.00%
1002	25000	5100	20.40%
1003	18000	4000	22.22%

I want Profit % for every order.

DAX calculates First row--> Store Result--> Second Row--> Store Result---.....Every Row gets its own result.

Business Usecases:

Create:

- GST
- Commission
- Profit %
- Discount Amount
- Net Profit
- Tax

Interview Question:

Which context is used in calculated columns?

Row Context

FILTER Context

Filter Context means DAX performs calculations only on the filtered data.

DAX calculates only the rows that match the current filters.

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Total Sales := SUM(Sales_Fact[Sales_Amount])

Interview Question:

Why does the same measure show different values in different pivot tables?

Because the measure is evaluated using the current Filter Context created by rows, columns, or report filters